**Zen AI Model Family**

# Zen-World

World Modeling, Physics Simulation, and Causal Reasoning at Scale

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**Abstract**

We present **Zen-World**, a 32-billion parameter world model capable of physics simulation, causal reasoning, counterfactual prediction, and long-horizon environment modeling. World models have emerged as a critical capability for planning-based AI systems: a model that can accurately predict how the world will evolve in response to actions enables agents to plan in imagination rather than through costly real-world trial and error. Zen-World achieves 87.3% on the PhysBench physical reasoning benchmark, 82.1% on CausalQA (causal question answering over dynamic scenes), and 91.4% physics consistency in a novel Physics Consistency Benchmark (PCB) we introduce. The model is built on the Zen MoDE architecture extended with a **Structured State Space World Representation (SSWR)**: a factored state representation that separates entities, their properties, and the physical and causal relations between them—enabling systematic generalization to novel physical configurations not seen during training.

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# 1 Introduction

A world model is a model that answers the question: *given the current state of the world and an action, what happens next?* This deceptively simple formulation encompasses some of the deepest challenges in AI: physical intuition, causal reasoning, object permanence, and counterfactual prediction.

World models serve as the foundation for model-based reinforcement learning: an agent with an accurate world model can plan multi-step action sequences entirely in mental simulation, dramatically improving sample efficiency and safety (by testing dangerous plans in imagination before executing them in reality).

Zen-World is designed as a general-purpose world model with three primary capabilities:

1. **Physics simulation**: Predict the physical evolution of scenes—trajectories, collisions, fluid dynamics, rigid body motion—from visual observations.
2. **Causal reasoning**: Answer questions about causal structure ("Why did X happen?", "Would Y have happened if Z had not occurred?").
3. **Counterfactual prediction**: Simulate counterfactual scenarios ("What would have happened if the red ball had been heavier?").

## 1.1 Model Overview

| Property | Value |
|---|---|
| Parameters | 32B |
| Architecture | Zen MoDE 32B + SSWR module |
| Context Length | 128K tokens (supports 10+ minute scene sequences) |
| Physics Timestep | Configurable 10ms–1s |
| Simulation Domains | Rigid body, fluid, cloth, soft body, molecular |
| Counterfactual Depth | Up to 50 intervention steps |
| Training Data | 2.1B simulated frames, 180M real-world video frames |

Table 1: Zen-World Model Specifications

# 2 Architecture

## 2.1 Structured State Space World Representation (SSWR)

The key architectural innovation in Zen-World is the SSWR: a factored representation of world state that separates four orthogonal components:

**Entity set** $\mathcal{E} = \{e_1, \ldots, e_N\}$: The set of objects in the scene, each with a learned embedding encoding its identity, material, shape, and learned physical properties.

**State tensor** $S \in \mathbb{R}^{N \times D_s}$: The current state of each entity: position, velocity, angular velocity, deformation fields.

**Relation graph** $\mathcal{R} \subseteq \mathcal{E} \times \mathcal{E} \times R$: Typed edges encoding physical relations: contact, attachment, fluid interaction, gravitational coupling.

**Intervention set** $\mathcal{I}$: Pending actions or counterfactual interventions to be applied at the next simulation step.

The world model's forward pass operates over this structured representation:

$$S_{t+1} = f_\phi(S_t, \mathcal{R}_t, \mathcal{I}_t) \tag{1}$$

$$\mathcal{R}_{t+1} = g_\phi(S_{t+1}, \mathcal{E}) \tag{2}$$

where $f_\phi$ is the dynamics model and $g_\phi$ updates the relation graph based on the new entity states (e.g., detecting new contacts).

The factored representation enables systematic generalization: new entities can be added to the simulation, and new relation types can be composed from known primitives.

### 2.2 Dynamics Model Architecture

The dynamics model $f_\phi$ is a message-passing graph neural network (GNN) operating over the SSWR graph, followed by 12 layers of the Zen MoDE transformer for global context integration:

$$m_{ij} = \mathrm{MLP}_e([h_i, h_j, r_{ij}, \Delta t]) \tag{3}$$

$$h'_i = h_i + \mathrm{MLP}_v\left(h_i, \sum_{j \in \mathcal{N}(i)} m_{ij}\right) \tag{4}$$

The GNN processes local physical interactions (contacts, collisions) efficiently, while the transformer handles long-range dependencies (e.g., a distant force field affecting multiple objects).

### 2.3 Causal Inference Module

For causal and counterfactual reasoning, Zen-World maintains a **causal graph** $\mathcal{C}$ alongside the physical simulation. The causal graph tracks which entities and actions causally contributed to each observed outcome.

Counterfactual queries ("What if X had not happened?") are answered by:

1. Identifying the intervention point in the causal graph.
2. Rolling back the simulation to just before the intervention.
3. Re-simulating with the modified initial conditions.
4. Comparing the counterfactual trajectory to the observed trajectory.

This approach is grounded in Pearl's do-calculus [1]: $P(\text{outcome} \mid \text{do}(\text{intervention}))$ is approximated by direct simulation.

## 3 Training

### 3.1 Dataset

| **Source** | **Frames** | **Proportion** | **Domain** |
|---|---|---|---|
| MuJoCo simulation | 800M | 38.1% | Rigid body, robotics |
| PhysX synthetic | 600M | 28.6% | General physics |
| Fluid simulation (SPH) | 300M | 14.3% | Fluid dynamics |
| Molecular dynamics | 200M | 9.5% | Molecular simulation |
| Real-world video (curated) | 180M | 8.6% | Physical events |
| Cloth/soft body sim. | 20M | 1.0% | Deformable objects |
| **Total** | **2,100,000,000** | 100% | |

Table 2: Zen-World Training Data Composition

### 3.2 Training Protocol

**Stage 1 – Physics encoder pretraining** (200K steps): The visual encoder + SSWR extraction network is trained to infer entity states and relation graphs from observations. Supervised by ground-truth simulation state (position, velocity, contact forces).

**Stage 2 – Forward dynamics training** (300K steps): The dynamics model is trained to predict future states from current states and actions. Loss is a combination of state prediction MSE and a physics-consistency regularizer (energy conservation, momentum conservation).

**Stage 3 – Causal graph learning** (100K steps): The causal inference module is trained on simulation traces with annotated causal graphs, learning to identify and propagate causal attributions.

**Stage 4 – Counterfactual fine-tuning** (50K steps): Fine-tuned on paired (observed, counterfactual) simulation trajectories generated by systematic intervention sampling.

| Stage | Steps | Batch | LR | Hardware |
|---|---|---|---|---|
| Physics encoder | 200K | 256 | 3e-4 | 64×A100 |
| Forward dynamics | 300K | 128 | 1e-4 | 128×A100 |
| Causal learning | 100K | 64 | 5e-5 | 32×A100 |
| Counterfactual FT | 50K | 32 | 2e-5 | 16×A100 |

Table 3: Zen-World Training Configuration

## 4 Evaluation

### 4.1 PhysBench

PhysBench evaluates physical reasoning across 5 categories: projectile motion, fluid dynamics, rigid body collisions, elastic deformation, and gravitational interactions.

| Model | Projectile | Fluid | Collision | Elastic | Mean |
|---|---|---|---|---|---|
| GPT-4o (text) | 73.2 | 61.4 | 78.3 | 64.1 | 69.3 |
| Claude 3.5 Sonnet | 76.8 | 65.2 | 81.4 | 68.3 | 72.9 |
| DreamerV3 | 68.3 | 71.4 | 74.2 | 72.1 | 71.5 |
| PHYRE | 79.4 | – | 82.3 | – | – |
| **Zen-World** | **89.2** | **83.4** | **91.7** | **84.8** | **87.3** |

Table 4: PhysBench Results (%, higher is better)

### 4.2 CausalQA

CausalQA presents video clips of physical events with questions about causal structure: "What caused the glass to fall?", "Would the ball have reached the target if the ramp had been steeper?"

| Model | Causal ID | Counterfactual | Intervention | Mean |
|---|---|---|---|---|
| GPT-4o + video | 74.3 | 61.2 | 68.4 | 68.0 |
| Gemini 1.5 Pro | 76.1 | 63.4 | 70.2 | 69.9 |
| CLEVRER-Base | 78.4 | 67.2 | 72.1 | 72.6 |
| **Zen-World** | **84.2** | **79.8** | **82.3** | **82.1** |

Table 5: CausalQA Results (%, higher is better)

The counterfactual category shows the largest gap (79.8% vs. 67.2%), confirming that Zen-World's explicit causal graph and simulation rollback provide a meaningful advantage over purely language-based approaches.

### 4.3 Physics Consistency Benchmark (PCB)

PCB is a novel benchmark we introduce: given a simulated trajectory, does the model's prediction satisfy physical conservation laws? We test energy conservation, momentum conservation, and angular momentum conservation in isolated systems.

| Model | Energy | Momentum | Ang. Mom. | Mean |
|---|---|---|---|---|
| Video prediction LSTM | 67.3 | 71.2 | 63.4 | 67.3 |
| DreamerV3 | 78.4 | 82.1 | 74.3 | 78.3 |
| **Zen-World** | **92.1** | **93.4** | **88.7** | **91.4** |

Table 6: Physics Consistency Benchmark (% of predictions satisfying conservation laws)

### 4.4 Long-Horizon Prediction

We evaluate simulation accuracy as a function of prediction horizon (seconds ahead):

| Horizon | Zen-World | DreamerV3 | RSSM |
|---|---|---|---|
| 0.5s | 94.2% | 91.3% | 87.4% |
| 2.0s | 88.7% | 79.2% | 71.3% |
| 10.0s | 76.4% | 58.1% | 44.2% |
| 60.0s | 61.3% | 31.4% | 18.7% |

Table 7: Long-Horizon Prediction Accuracy (% states within 5% error)

## 5 Applications

### 5.1 Model-Based Planning for Robotics

Zen-World serves as the environment model in the Hanzo robotics planning stack. Given a robot's sensor observations, the model maintains an internal simulation of the scene and allows the motion planner to evaluate candidate action sequences entirely in simulation before committing to execution, reducing physical trial attempts by 84%.

### 5.2 Scientific Simulation

Zen-World's molecular dynamics domain enables protein folding stability prediction and drug binding affinity estimation by simulating molecular-scale physical interactions.

### 5.3 Autonomous Driving Safety

In autonomous driving contexts, Zen-World is used to simulate counterfactual accident scenarios: "Would the collision have been avoided if the vehicle had braked 0.5s earlier?" This enables post-incident analysis and safety system certification without physical testing.

## 6 Integration

```python
from zen import ZenWorld
import numpy as np

model = ZenWorld.from_pretrained("zenlm/zen-world-32b")

# Parse scene from image
scene = model.parse_scene("scene.jpg")
print(f"Detected {len(scene.entities)} objects")

# Simulate 5 seconds forward
trajectory = model.simulate(
    scene=scene,
    action={"push_object": "ball_1", "force": [2.0, 0.0, 0.0]},
    duration_sec=5.0,
    timestep_ms=50
)

# Counterfactual query
cf_trajectory = model.counterfactual(
    trajectory=trajectory,
    intervention={"at_time": 0.5, "change_mass": {"entity": "ball_1", "
        new_mass": 5.0}}
)

print(f"Original final position: {trajectory.final_state.entities['
    ball_1'].position}")
print(f"Counterfactual final: {cf_trajectory.final_state.entities['
    ball_1'].position}")
```

Listing 1: Zen-World Physics Simulation

## 7 Related Work

RSSM [2] introduced latent dynamics models for model-based RL. DreamerV3 [3] demonstrated general applicability across diverse domains. PHYRE [4] established physical reasoning benchmarks. CLEVRER [5] introduced causal and counterfactual video question answering. Zen-World advances this line by combining structured SSWR representations with 32B Zen MoDE scale, achieving state-of-the-art results across physical reasoning, causal QA, and physics consistency metrics.

## 8 Conclusion

Zen-World demonstrates that world modeling at 32B scale with structured state representations (SSWR), explicit causal graph tracking, and simulation-based counterfactual reasoning produces substantially better physical understanding than purely learned end-to-end predictors. The 87.3% PhysBench, 82.1% CausalQA, and 91.4% PCB results represent the current state of the art across all three evaluation dimensions, with particularly strong gains in counterfactual prediction—the capability most critical for planning and scientific analysis.

## References


[1] J. Pearl, "Causality: Models, Reasoning, and Inference," 2nd ed., Cambridge University Press, 2009.

[2] D. Hafner et al., "Learning Latent Dynamics for Planning from Pixels," ICML, 2019.

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[4] A. Bakhtin et al., "PHYRE: A New Evaluation Framework for Physical Reasoning," NeurIPS, 2019.

[5] K. Yi et al., "CLEVRER: Collision Events for Video Representation and Reasoning," ICLR, 2020.